

NCERT Practical Assignment #01

Ubuntu Installation, Nginx Deployment & NCERT Front-End Website

Course: Advanced Cyber Security

Organization: National Cyber Emergency Response Team (NCERT)

Date: 02 July 2026

Deadline: Today (End of Lab Session)

Total Marks: 100

Objective

The objective of this assignment is to:

- Install the latest stable version of Ubuntu Linux.
 - Install the latest version of the Nginx web server.
 - Develop a professional front-end website for the National Cyber Emergency Response Team (NCERT).
 - Deploy the website using Nginx.
 - Make the website accessible on <http://localhost:8080>.
 - Gain an understanding of the major operational domains within a Cyber Emergency Response Team.
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Assignment Tasks

Task 1 – Ubuntu Installation

Install the latest stable version of **Ubuntu Linux** using any virtualization platform (VirtualBox, VMware, Hyper-V) or on a physical machine.

Ensure the operating system is fully updated.

Task 2 – Nginx Installation

Install the latest stable version of **Nginx** on Ubuntu.

Verify that the web server is running successfully.

Task 3 – Website Deployment

Configure and deploy your website using **Nginx**.

The website **must be accessible at:**

<http://localhost:8080>

Task 4 – Develop an NCERT Welcome Page

Create a professional and responsive front-end landing page for the **National Cyber Emergency Response Team (NCERT)**.

Your homepage should include:

- NCERT Logo
 - Website title
 - Navigation menu
 - Professional banner (Hero Section)
 - Attractive layout
 - Responsive design
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Task 5 – NCERT Departments

Create separate sections explaining the following departments and their responsibilities.

1. Cyber Security

Include information about:

- Network Security
- Endpoint Security
- Vulnerability Assessment
- Penetration Testing
- Threat Intelligence

- Security Awareness
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2. Security Operations Center (SOC)

Explain:

- 24/7 Security Monitoring
 - SIEM
 - Incident Detection
 - Log Monitoring
 - Threat Hunting
 - Incident Response
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3. Software Development

Explain:

- Secure Software Development
 - Web Applications
 - Mobile Applications
 - API Development
 - DevSecOps
 - Secure Coding Practices
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4. Governance, Risk & Compliance (GRC)

Explain:

- Information Security Policies
- Risk Assessment
- Compliance Management
- ISO 27001

- NIST Cybersecurity Framework
 - Internal Security Audits
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5. Digital Forensics Laboratory

Explain:

- Computer Forensics
 - Mobile Device Forensics
 - Memory Forensics
 - Disk Analysis
 - Malware Analysis
 - Digital Evidence Collection
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Task 6 – How NCERT Functions

Create a section explaining the workflow of a Cyber Emergency Response Team.

Your explanation should include the following stages:

- Threat Detection
- Alert Generation
- Incident Analysis
- Containment
- Eradication
- Recovery
- Reporting
- Lessons Learned

Provide a brief description of each stage.

Task 7 – Website Design Requirements

Your website should include:

- Professional user interface
- Responsive design
- Header and navigation bar
- Hero section
- Service cards
- Icons
- Images
- Footer
- Contact section
- Professional color theme

Students may use:

- HTML5
- CSS3
- Bootstrap
- JavaScript

Task 8 – Footer

Include the following information in the footer:

- National Cyber Emergency Response Team (NCERT)
- © 2026 NCERT
- Developed by: **Your Name**
- Student ID
- Course Name

Submission Requirements

Each student must submit:

- Ubuntu version
- Nginx version
- Complete website project folder
- Screenshots of:
 - Ubuntu desktop
 - Installed Nginx version
 - Website running successfully on <http://localhost:8080>
- A brief report (1–2 pages) describing:
 - What they learned
 - Challenges faced
 - How they resolved those challenges

Evaluation Criteria

Criteria	Marks
Ubuntu Installation	10
Latest Nginx Installation	10
Website Deployment on Port 8080	10
Website Design & User Interface	20
Content Accuracy (NCERT Services)	20
Responsiveness & Layout	10
Creativity & Professional Appearance	10
Documentation & Submission	10
Total	100 Marks

Learning Outcomes

After completing this assignment, students should be able to:

- Install and manage Ubuntu Linux.
- Install and deploy applications using the Nginx web server.
- Host a website locally using Nginx.
- Understand the core functions of the National Cyber Emergency Response Team (NCERT).
- Design and develop a professional, responsive front-end website.
- Present cybersecurity services in a structured and visually appealing manner.

Submission Deadline: Today before the end of the lab session. Students should be prepared to demonstrate their website live during the practical evaluation